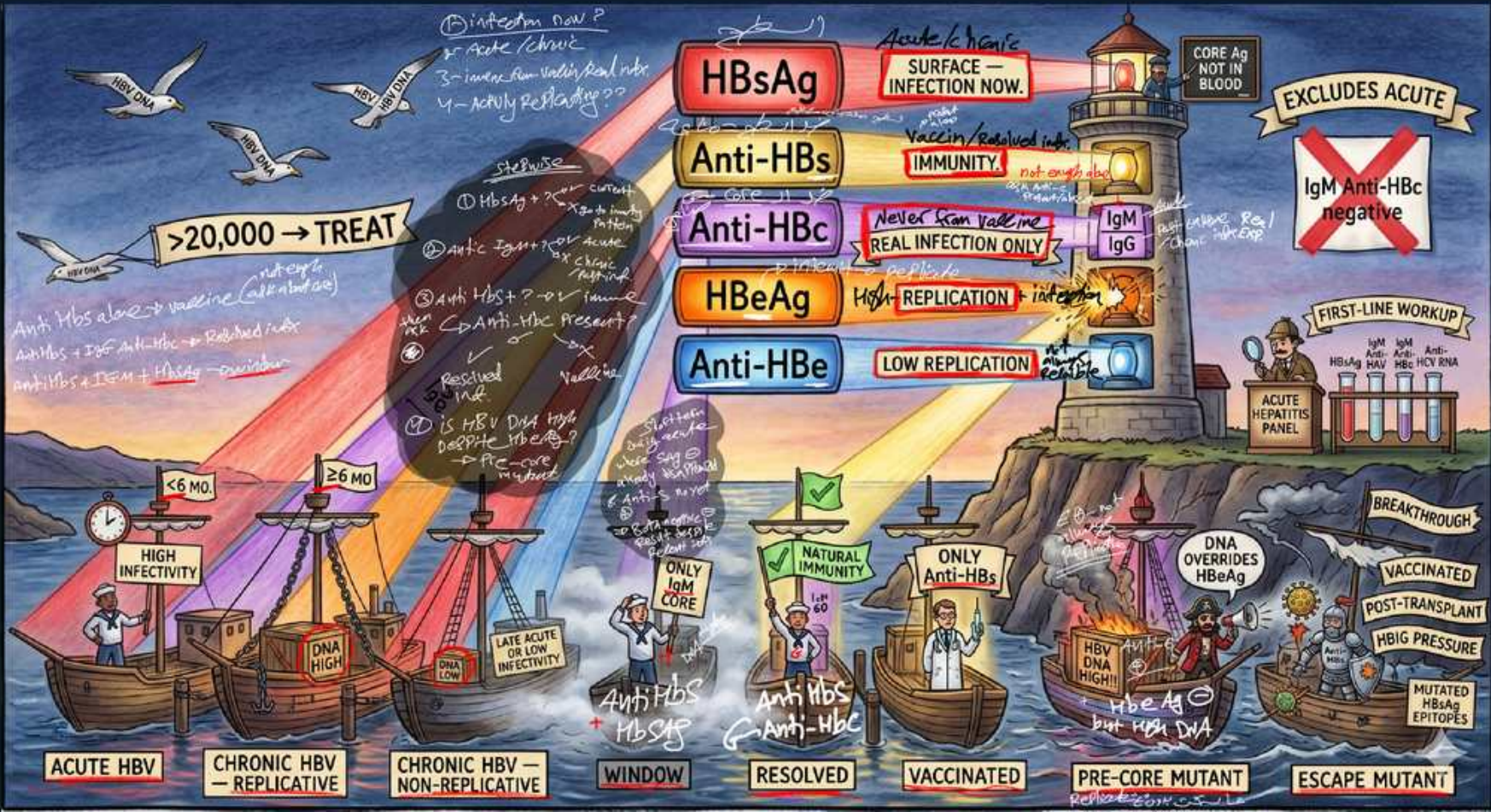


Viral Hepatitis — HBV Serology Interpretation



1. HBsAg — active infection

WHAT YOU SEE

Red HBsAg beam banner: 'SURFACE — INFECTION NOW'

WHAT IT ENCODES

HBsAg (surface antigen) positive = current active HBV infection, acute or chronic, with virus/antigen present in blood. It is the FIRST serologic marker to appear (1-10 weeks after exposure, before symptoms) and the screening test for active infection. It is the antigen the vaccine is made from (recombinant HBsAg).

BOARD TRAP

HBsAg persisting >6 months DEFINES chronic infection. A common trap: HBsAg can be transiently positive for up to ~3 weeks after HBV vaccination — but the vaccine never produces anti-HBc, which distinguishes it from real infection.

2. Anti-HBs — immunity

WHAT YOU SEE

Green Anti-HBs banner: 'IMMUNITY'

WHAT IT ENCODES

Anti-HBs (surface antibody) = IMMUNITY/protection, arising either from resolved natural infection OR from vaccination. A titer ≥ 10 mIU/mL is considered protective. Its appearance after the window period marks recovery and non-infectivity.

BOARD TRAP

Anti-HBs is the ONLY HBV marker present after vaccination. If a stem shows anti-HBs positive but anti-HBc NEGATIVE, the patient is vaccinated, not recovered from infection.

3. Anti-HBc — real infection only

WHAT YOU SEE

Purple Anti-HBc banner 'REAL INFECTION ONLY' with IgM/IgG

WHAT IT ENCODES

Anti-HBc (core antibody) appears only with true HBV infection — NEVER from vaccine (the vaccine contains no core antigen). IgM anti-HBc = acute/recent infection (and is the sole marker during the window period); IgG anti-HBc = past or chronic infection and persists for life.

BOARD TRAP

Anti-HBc is the marker that distinguishes vaccinated (anti-HBc NEGATIVE) from naturally resolved infection (anti-HBc POSITIVE). 'Isolated anti-HBc positive' = window period, distant resolved infection with waned anti-HBs, or false positive — order HBV DNA to sort it out.

4. HBeAg — high replication

WHAT YOU SEE

Orange HBeAg banner 'HIGH REPLICATION / INFECTIOUS'

WHAT IT ENCODES

HBeAg (envelope/secreted antigen) = active viral replication, high HBV DNA, and high infectivity. Its presence in a chronic carrier marks the immune-active, treatment-relevant, highly contagious phase and the highest perinatal transmission risk.

BOARD TRAP

HBeAg positivity = highly contagious; an HBeAg+ pregnant mother confers very high perinatal transmission, mandating HBIG + vaccine to the neonate. Don't equate HBeAg-negative with low DNA — see the pre-core mutant.

5. Anti-HBe — low replication

WHAT YOU SEE

Blue Anti-HBe banner 'LOW REPLICATION'

WHAT IT ENCODES

Anti-HBe appears after HBeAg seroconversion and generally signals declining replication, lower HBV DNA, and reduced infectivity — often the transition to the inactive carrier state.

BOARD TRAP

Anti-HBe usually means low replication, BUT a pre-core/basal-core promoter mutant can be anti-HBe positive yet still have HIGH HBV DNA and active disease. Always confirm with HBV DNA before calling a patient 'inactive.'

6. IgM anti-HBc excludes acute

WHAT YOU SEE

Top-right red-crossed box: 'EXCLUDES ACUTE — IgM anti-HBc negative'

WHAT IT ENCODES

IgM anti-HBc is the marker of ACUTE/recent HBV: positive confirms acute infection, negative essentially rules out acute infection. It is also the ONLY positive marker during the window period when HBsAg has cleared but anti-HBs has not yet risen.

BOARD TRAP

In the window period, IgM anti-HBc may be the SOLE positive marker (HBsAg negative, anti-HBs negative) — so a negative HBsAg does not exclude recent acute HBV. Always include core antibody, not just HBsAg, when working up acute hepatitis.

7. Acute HBV ship

WHAT YOU SEE

Leftmost ship labeled ACUTE HBV (<6 MO, high infectivity)

WHAT IT ENCODES

Acute HBV profile: HBsAg+, IgM anti-HBc+, HBeAg+ (high DNA, high infectivity). In immunocompetent adults >95% clear spontaneously within ~6 months, developing anti-HBs and IgG anti-HBc.

BOARD TRAP

Duration defines outcome: <6 months = acute; HBsAg persistence beyond 6 months = chronic. Acute HBV in a healthy adult is usually NOT treated with antivirals (it self-resolves) — treatment is reserved for severe/fulminant or chronic disease.

8. Chronic HBV — replicative

WHAT YOU SEE

Second ship CHRONIC HBV — REPLICATIVE (>6 MO)

WHAT IT ENCODES

Chronic replicative (immune-active) HBV: HBsAg+ >6 months, HBeAg+, HIGH HBV DNA, IgG anti-HBc+, and elevated ALT. This phase carries the highest risk of progression and is the treatment-eligible group — first-line agents are entecavir, tenofovir (TDF/TAF), or pegylated interferon.

BOARD TRAP

HBeAg+ chronic with high DNA + elevated ALT = treat. Don't withhold therapy here. Conversely, don't treat the immune-tolerant phase (HBeAg+, high DNA, but persistently normal ALT) reflexively — ALT and fibrosis guide treatment timing.

9. Chronic HBV — non-replicative

WHAT YOU SEE

Third ship CHRONIC HBV — NON-REPLICATIVE

WHAT IT ENCODES

Inactive carrier (non-replicative): HBsAg+, HBeAg negative, anti-HBe+, low/undetectable HBV DNA (<2000 IU/mL), and persistently normal ALT. Usually monitored rather than treated, with periodic ALT/DNA and HCC surveillance.

BOARD TRAP

Low DNA + normal ALT = surveillance, not antivirals. BUT an inactive carrier can reactivate (e.g., with chemo, rituximab, steroids) — give prophylactic entecavir/tenofovir before immunosuppression, since HBsAg+ patients can reactivate fatally.

10. Window period ship

WHAT YOU SEE

Ship labeled WINDOW with 'ONLY anti-HBc / IgM core'

WHAT IT ENCODES

Window period: HBsAg has disappeared but anti-HBs has not yet risen, so the ONLY positive serology is anti-HBc (IgM). It represents the gap between clearance of antigen and detectable protective antibody.

BOARD TRAP

THE classic exam trap: a patient with acute hepatitis whose HBsAg AND anti-HBs are both negative — diagnosis hinges on the isolated IgM anti-HBc. Missing the window period leads to wrongly excluding HBV.

11. Resolved ship

WHAT YOU SEE

Ship RESOLVED with green check 'NATURAL IMMUNITY, anti-HBs + anti-HBc'

WHAT IT ENCODES

Resolved (past) infection with natural immunity: anti-HBs+ AND IgG anti-HBc+, with HBsAg negative. Both antibodies present = the body cleared a real infection and is now immune.

BOARD TRAP

Both anti-HBs and anti-HBc positive = PAST INFECTION, not vaccination. The presence of anti-HBc is the tell — vaccine never makes anti-HBc.

12. Vaccinated ship

WHAT YOU SEE

Ship VACCINATED 'ONLY Anti-HBs'

WHAT IT ENCODES

Vaccinated profile: anti-HBs+ ONLY (HBsAg negative, anti-HBc negative). The recombinant HBsAg vaccine generates protective surface antibody without ever exposing the immune system to core antigen.

BOARD TRAP

Isolated anti-HBs = vaccinated; add a positive anti-HBc and it becomes resolved natural infection. The single discriminator between vaccine immunity and recovered immunity is anti-HBc.

13. Pre-core mutant ship

WHAT YOU SEE

Ship PRE-CORE MUTANT, fire/high DNA, 'DNA OVERRIDES HBeAg'

WHAT IT ENCODES

Pre-core / basal-core promoter mutant: a strain unable to make HBeAg, so it is HBeAg-NEGATIVE and anti-HBe-positive yet still has HIGH HBV DNA and active necroinflammatory disease (elevated ALT). It is treatment-eligible like other active chronic HBV.

BOARD TRAP

Don't be fooled by negative HBeAg into calling the patient inactive — HBV DNA OVERRIDES HBeAg status. HBeAg-negative chronic hepatitis B with high DNA and elevated ALT still needs antiviral therapy.